

Sistemas Distribuídos

Licenciatura em Engenharia de Telecomunicações e Informática

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Version 1

Assignment 1

Background

In the Projeto de Software course (PSOFT), students developed a RESTful application known as the AISafe Flight Management System. The system provides a set of REST APIs for the management of core domain entities, namely:

- Aircraft (both models and individual instances)
- Airports
- Flight routes
- Maintenance records and associated templates
- Scheduled flight operations

Motivation

Although the application meets its functional requirements, its monolithic architecture imposes clear limitations on several essential properties of distributed systems. In particular, horizontal scalability, fault tolerance and overall resilience remain constrained. A centralised design inevitably introduces single points of failure, performance degradation under concurrent load, and difficulties in preserving availability when individual nodes fail or network partitions occur.

Objectives

The main goal of this assignment is to redesign the AISafe Flight Management System according to established principles of distributed computing. Emphasis is placed on decentralisation, interoperability and robustness in the presence of failures.

Three complementary distribution strategies must be realised:

1. Domain-driven segregation

The monolithic application is to be decomposed into three distinct yet collaborating services that execute across a network. Communication between services shall rely on lightweight protocols (HTTP/REST being the expected choice) in order to achieve loose coupling, independent deployability and improved long-term maintainability.

A possible domain partition (to be refined according to group size) is the following:

- Aircraft & Maintenance service – management of aircraft models and instances, operational status, maintenance templates and records;
- Airports & Routes service – registration and querying of airports, runway and certification data, route definition and history;
- Flight Operations service – scheduling of flights, assignment of aircraft to routes, real-time availability checks and operational queries.

2. Instance replication

Multiple replicas of each service must be deployed so that high availability and load distribution can be achieved. When an HTTP GET request reaches a replica that does not hold the required data (owing to data partitioning or sharding), that replica is expected to forward the query to its peers in a peer-to-peer manner, aggregate the collected responses and return a coherent answer to the client. The resulting design should favour eventual consistency, redundancy-based fault tolerance and resilience to partial network failures, thereby aligning with the availability and partition-tolerance priorities of the CAP theorem.

3. Security considerations for operational aviation data

Fleet, route and maintenance information carries operational sensitivity. Consequently, the distributed realisation must preserve data integrity, confidentiality of sensitive operational records and sound security practices across all services and communication channels. Concrete measures to be addressed include:

- Inter-service authentication
- Encryption of data in transit
- Access control and audit logging

Organization

- Team composition: 3 students per group (exceptions allowed for 4 or 2).
- Requirements are tailored according to group size to ensure equitable workload distribution.

Expected Deliverables (summary):

- Working distributed implementation of the three services with replication
- Demonstration of peer-to-peer query propagation and response aggregation
- Evidence of inter-service authentication, encryption and role-based access control
- Documentation of design decisions, consistency model and fault-tolerance mechanisms
- Source code and deployment instructions

Assessment:

- Mandatory
- Submission deadline: 2026-10-25
- Mode: Individual, synchronous, in-person
- Weight: 20%